

Phase 1 - Week 3 - Lesson Day 3

Descriptive Statistics

Today's Objectives

Charts show you the shape of data visually; descriptive statistics summarize that same shape using numbers. Together they form the two halves of exploring any new dataset. Today's concepts are the foundation for everything statistical later in the course, including hypothesis testing and model evaluation.

- Calculate and interpret mean, median, and mode
- Calculate and interpret variance and standard deviation
- Understand what a distribution is and recognize common shapes

1. Mean, Median, and Mode

Definition: These are the three measures of "central tendency" - different ways of answering "what's a typical value in this data?"

Mean - the average: add up every value, divide by how many there are.

Median - the middle value when all values are sorted. If there's an even count, it's the average of the two middle values.

Mode - the value that appears most often.

```
import numpy as np
from scipy import stats

data = np.array([10, 12, 12, 15, 18, 95]) # notice the 95 is a clear outlier

print(np.mean(data))      # 27.0 - heavily pulled upward by the outlier (95)
print(np.median(data))    # 13.5 - barely affected by the outlier
print(stats.mode(data, keepdims=True).mode[0]) # 12 - the most frequent value
```

Why this matters: the mean is sensitive to outliers - a single extreme value can pull it far from what feels "typical." The median is far more robust to outliers, which is why you'll often see median household income reported instead of mean income (a few billionaires would distort the mean dramatically).

2. Variance and Standard Deviation

Definition: These measure how *spread out* the data is around the mean - not just "what's typical," but "how much do values vary from that typical value?"

```
print(np.var(data))      # variance - average of the squared differences from the mean
print(np.std(data))      # standard deviation - the square root of variance
```

Why standard deviation, not just variance? Variance is in squared units (e.g. squared dollars), which is hard to interpret intuitively. Standard deviation takes the square root, bringing it back into the original units - a standard deviation of 10 points on a test is directly comparable to the scores themselves, while a variance of 100 (points squared) is not.

Rule of thumb: a small standard deviation means values cluster tightly around the mean (consistent data). A large standard deviation means values are spread widely (inconsistent, more variable data).

3. Distributions

Definition: A *distribution* describes the overall shape of how values in a dataset are spread out - which values are common, which are rare, and the overall pattern. You already saw distributions visually with histograms on Day 1; today connects that shape to the numbers behind it.

Normal distribution - the classic symmetric "bell curve." Most values cluster near the mean, with fewer values further away in either direction. Many natural phenomena (heights, test scores, measurement errors) approximate this shape.

Skewed distribution - not symmetric. A *right-skewed* (positively skewed) distribution has a long tail stretching to the right (e.g. income - most people earn a moderate amount, a few earn vastly more). A *left-skewed* distribution has a long tail to the left.

```
import matplotlib.pyplot as plt

normal_data = np.random.normal(50, 10, 1000)    # symmetric bell curve
skewed_data = np.random.exponential(10, 1000)    # right-skewed (long tail to the right)

plt.hist(normal_data, bins=30, alpha=0.6, label="Normal")
plt.hist(skewed_data, bins=30, alpha=0.6, label="Right-skewed")
plt.legend()
plt.show()
```

Why this matters in practice: when mean and median are close together, the data is roughly symmetric. When the mean is noticeably higher than the median, the data is right-skewed (a few large outliers pull the mean up); when the mean is noticeably lower than the median, it's left-skewed. This simple comparison is a fast way to detect skew without even looking at a chart.

Today's Exercise

Open [day-3-descriptive-stats.ipynb](#). All exercises are fully solved, calculating statistics both manually (to build intuition) and with built-in functions (to confirm the manual calculation is correct).

Exercise Questions (5 total - all solved in the notebook)

1. Manual vs built-in calculation

Calculate the mean and standard deviation of a small array manually (writing out the formula), then verify with `np.mean()` and `np.std()`.

2. Mean vs median with an outlier

Create a dataset, calculate its mean and median, then add one extreme outlier and recalculate both - compare how much each one moved.

3. Find the mode

Create a dataset with a clearly repeated value and find its mode using `scipy.stats`.

4. Compare spread of two datasets

Create two datasets with the same mean but different standard deviations, and explain in a comment what that difference means practically.

5. Detect skew from mean vs median

Generate a right-skewed dataset (e.g. using `np.random.exponential`), calculate its mean and median, and explain in a comment what their relationship tells you about the skew.

Once you and Halimo have reviewed all five solved exercises, move on to Lesson Day 4 (Full EDA - combining everything from this week into one analysis).